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REVIEW ARTICLE

Science Democratization for Rigor, Relevance, and Resilience

Marlen Z. Gonzalez[1] | Rikki S. Laser[2] | Sara Catalina Carrillo[2] | Xinyi Deng[2] | Elizabeth Riley[2] | Riesa Y. Cassano-Coleman[3] | Zahra Baninameh[4] | Nora H Prior[2] | Gloria Chen[2] | Mary K. MacMillan[5] | Genevieve M Wager[6] | Sanaea Z. Bhagwagar[7] | Lisa Morton[8] | Samantha De Leon Sautu[2,9] | Sará Y. King[10] | Maureen Dunne[11] | Kuricheses Bone-Alexander[12] | Frances Colón[13] | Eve De Rosa[2] | Senegal Alfred Mabry[2]

[1]Executive Director of Community Neuroscience Initiative, Cornell University, Ithaca, New York, USA | [2]Department of Psychology, Cornell University, Ithaca, New York, USA | [3]Department of Brain and Cognitive Sciences, University of Rochester, Rochester, New York, USA | [4]Biomedical & Biological Sciences Ph.D. Program, Cornell University, Ithaca, New York, USA | [5]Department of Biomedical Sciences, University of Virginia, Charlottesville, Virginia, USA | [6]Department of Biomedical Engineering, Cornell University, Ithaca, New York, USA | [7]State University of New York Upstate Medical University, Syracuse, New York, USA | [8]Department of Clinical Psychology, Teachers College, Columbia University, New York, New York, USA | [9]National Secretary of Science and Technology (SENACYT), Panama City, Panama | [10]MindHeart, San Francisco, USA | [11]Faculty Affiliate, Discovery Partners Institute, University of Illinois, Champaign, Illinois, USA | [12]STEAM at Dr. King Elementary School, Syracuse, New York, USA | [13]Center for American Progress, Washington, District of Columbia, USA

Correspondence: Marlen Z. Gonzalez (mzg5@cornell.edu)

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ABSTRACT

Developmental psychobiology and neuroscience hold the promise to improve children's lives but also the peril to entrench marginalization when insights are misapplied or stripped of context. Diversification tilts us towards promise, but as political forces threaten inclusive research practices and public trust in science, developmental researchers face a critical moment. This paper argues that science democratization—grounded in care, inclusivity, and shared authority—can make our science more rigorous, relevant, and resilient. We begin by reviewing how gender, ethnoracial, and cognitive diversity among researchers and participants has expanded the field's reach and sharpened its questions. We then turn to democratization as a relational stance centering care and agency, with the enhancement of our science as a consequence. To ground this approach, we describe an illustrative gender-inclusion event led by the Community Neuroscience Initiative (CNI), which brought together scientists and community members for dialogue, shared learning, and collaboration. Finally, we offer readings, practical recommendations, and open questions for readers interested in applying these ideas to their own work. Written collaboratively with input from all stakeholders involved, this manuscript offers a timely vision for a more ethical, inclusive, and impactful developmental psychobiology and neuroscience.

1 | Introduction

The use of developmental psychobiology to inform public policy has been fraught with both promise and peril (Farah 2018). As developmental psychobiology draws from the related fields of neuroscience, psychology, and biology, it is marked by an interdisciplinary approach to understanding how young people develop emotionally, socially, and physiologically, which can inform practices that support healthy development[1]. On the one hand, our research has resulted in meaningful changes that align

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social and biological realities. For example, many high schools have moved to later start times in part due to research indicating that circadian rhythms and neural pruning require more and later sleep for teens in order to function properly (Tarokh et al. 2016). The result has been better grades, behavioral management, and mood for adolescents in high school (Wahlstrom et al. 2014). On the other hand, there are numerous examples where our research, either because of misinterpretation, misapplication, or lack of clear communication, has caused harm. For example, in the United Kingdom, research on infancy from developmental neuroscience was used to justify redirecting and skewing resources for families in poverty towards those with children under 3 years of age, at the expense of families with older children (Wastell and White 2012). In the United States, the risk of a negative impact on childhood policy is exacerbated along ethnoracial disparities: minoritized children are both overrepresented in poverty and underrepresented in the field of developmental psychobiology due to historical and systemic factors (Green et al. 2022; Heard-Garris et al. 2021).

Research in public and community health suggests that participatory research is one way of increasing rigor, generalizability, and positive impact (Lasker and Weiss 2003). Participatory research describes a host of methods that aim to (1) include diverse communities in local contexts in the governance of science and (2) promote solving specific community problems in the course of the research process. It is a framework that emphasizes the ethical coproduction of knowledge among diverse stakeholders (Bandola-Gill et al. 2023). To our knowledge, only one developmental psychobiology study has been published on applying concepts from community-based participatory research (CBPR) (La Scala et al. 2023). In their manuscript, the authors suggest specific tools (positionality maps and community advisory boards) that scientists can use to help bridge the gap in understanding, have immediate positive community impact, make power dynamics between researchers and community members more horizontal, and otherwise enlist community members in participant recruitment, implementation, quality, and impact of the work (La Scala et al. 2023). Others have made broader recommendations on how to diversify the participant pool in developmental neuroscience, calling for systemic changes including the sustained diversification of our workforce via recruitment, retention, and promotion strategies (Garcini et al. 2022).

However, sweeping changes in the US federal research funding environment put in peril the nascent strides by developmental psychobiologists to include participatory methods. The terrain for inclusive participatory work has never been particularly favorable in our field, but it looks even worse in 2025. Executive orders have decreed that the National Institutes of Health (NIH) and the National Science Foundation (NSF) should deter any and all work related to diversity, inclusion, and equity (DEI). Broader communications have also emphasized refraining from inclusive work on "abilities," potentially resulting in the isolation of individuals with disabilities and neurodivergent people (U.S. Office of the Attorney General 2025). At the same time, the federal government has tried to reduce indirect costs—which are essential for research institutions to keep their infrastructure—to 15%, impacting all scientific work. Under this forced scarcity, universities and researchers have responded by rolling back DEI initiatives and engaging in appeasement to avoid losing federal funds. While legal battles have begun and will continue regarding the legitimacy of these actions (Malakoff 2025), one thing is clear: some of us are faltering. In some cases, this may be due to values, but we suspect that many more are simply lacking tools and examples to help them advocate effectively for the value of inclusive practices while their jobs and work are on the line. This may be especially true for trainees and others in vulnerable positions.

Therefore, this manuscript aims to provide individuals with assurance, philosophy, and practical examples of why diversity and community-engaged work are always important, but especially when science is under attack. Specifically, we will discuss this within the context of the Community Neuroscience Initiative (CNI), a project at Cornell University under the Bronfenbrenner Center for Translational Research (BCTR). This article is written by members of the CNI, trainees, and nonacademic community members. The CNI's mission is to "democratize neuroscience." In brief, science democratization is a philosophical stance on the rights and responsibilities of scientists and the public in the creation, dissemination, and application of science, which will be discussed in more depth later. What follows in this article is (1) a short review of the ways diversity and inclusion impact innovation, (2) a discussion on science democratization and what it means for our field, (3) an example of democratization in practice focused on a specific CNI gender-inclusive event as experienced by the writers of this paper, and (4) further reading and practices for the curious. If this article achieves anything, we hope it gives researchers at different stages of their careers courage. We intend to provide language and examples needed to understand how centering democratization of science, which by its nature requires representational diversity, can increase the rigor, public relevance, and resilience of our work.

2 | Diversity and Discovery

In *Regents of the University of California v. Bakke* (1978), the Supreme Court ruled that while quotas were impermissible, student body diversity promoted a "vigorous exchange of ideas" beneficial to higher education and society. That principle is now under threat (U.S. Office of the Attorney General 2025), prompting a renewed question: has diversity helped science? While a full review is beyond our scope, the evidence is clear—diversity, when achieved, drives discovery. Here, we highlight how gender, ethnoracial, and neurodiversity among researchers, as well as participant inclusion, have expanded the epistemic and methodological reach of developmental psychobiology. This is not to suggest that diversity must "prove" its utility; in a democratic society, access to the production of knowledge is a good in itself (Irzik and Kurtulmus 2024). Still, it bears repeating: diversity and discovery coincide.

2.1 | Diversity Among Researchers

2.1.1 | Gender Diversity Improves Team Performance

The biggest diversity gains in psychobiology and neuroscience have been in increasing the representation of women. Women

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make up over 50% of undergraduate and graduate students in neuroscience, yet only 31% of tenure-track faculty and 14% of full professors (Machlovi et al. 2021). This is more than a pipeline issue—it affects the quality and scope of scientific discovery. Gender-diverse teams demonstrate greater collective intelligence, broader problem-solving approaches, and more innovative research questions (Riedl et al. 2021). The exact mechanisms by which diversity in leadership mediates success are unknown, but proposed mechanisms include better identification of novel and timely research questions (Gomez and Bernet 2019; Yang et al. 2022), better use of various methodologies (Martins and Sohn 2022), and more accurate interpretations of data (Levine et al. 2014), thereby enriching the field's intellectual landscape. One study showed that women play a direct role in impacting collective intelligence factors within a group via greater perceptive abilities and the ability to foster more collective collaboration (Woolley et al. 2010). In other words, teams that share power more equitably among diverse individuals benefit from improved problem-solving abilities and greater creativity. Increasing equitable representation at the highest levels can expand scientific inquiry and increase the field's capacity to address complex challenges effectively.

2.1.2 | Ethnoracial Diversity Improves Impact and Participation

Compared to gender diversity, ethnoracial diversity has lagged behind in psychobiology. Black and Latine/x individuals make up 30% of the U.S. population, yet remain underrepresented across all facets of neuroscience. A recent survey of PhD psychology programs found that among subfields using neuroscientific (psychobiological) techniques, the racial/ethnic breakdown was 9% Latine/x, 4% Black, 13% Asian, and 53% White, with 18% selecting "unknown" (American Psychological Association 2024). Faculty representation is similarly disproportionate, with only 11% of faculty positions held by Black or Latine/x scholars (Berger-Sweeney 2023). This is unfortunate because ethnoracial diversity significantly strengthens scientific impact. A large-scale analysis of general academic publishing found that papers authored by more ethnically diverse teams had higher citation rates and greater impact factors for researchers (AlShebli et al. 2018). Outside academia, there is some research suggesting that ethnoracial diversity at the county level correlates with less stereotypical implicit bias towards other minorities' identities, including sexual orientation and disability (Tiv and Spence 2024). Although these general results may not always translate to academic settings, evidence suggests that ethnoracial diversity within a broader context may either be a good indicator of other forms of inclusion or foster an inclusion culture. Within academia, there is additional evidence that ethnoracial diversity also broadens research topics and increases participant diversity in psychological science. In a review of published articles on race, researchers found that ethnoracially minoritized researchers are more likely to investigate how race influences developmental, cognitive, and social psychological processes, expanding the scope of the field (Roberts et al. 2020). This diversification of both topics of study ensures that psychological science reflects the lived experiences of a broader range of populations (Roberts and Mortenson 2023).

2.1.3 | Neurodiversity Is Associated With Traits Important to Scientific Research

Neurodiversity—cognitive variation across individuals—is an often invisible contributor to science. Neurodiversity recognizes the natural variation in human cognition and how different cognitive styles contribute to scientific progress and society (Dwyer et al. 2023; Syharat et al. 2023). While often framed in clinical terms (e.g., attention deficit disorder, ADHD, autism spectrum), neurodiversity encompasses a range of traits that can enhance attention to detail, creativity, or pattern recognition. These traits can be assets in research when supported by inclusive structures (Syharat et al. 2023). Approaches like Universal Design for Learning (UDL) aim to create environments that provide these needed supports across the neurodiversity spectrum for the benefit of all (Boothe et al. 2018). For example, multisensory learning like closed captions is useful for the hearing impaired, individuals with ADHD, and individuals on the autism spectrum, but it also benefits the general public (Shams and Seitz 2008). This illustrates how innovations focused on neurodiversity inclusion end up benefiting all.

2.1.4 | Intersectional Identities Are Important to Consider

Intersectionality refers to a framework for revealing power dynamics and understanding how oppression and privilege across multiple social identities (e.g., socioeconomic status, gender, race, and disability) in a single individual can impact their experiences. This theory, developed by legal scholar Kimberle Crenshaw, has traversed disciplines, including psychological science (Rutherford and Davidson 2019). In neuroscience, a recent paper focused on the need for researchers to consider intersectional identities and especially compounding oppressions for participants in neuroimaging research (Dhamala et al. 2025). Other work has pushed back on the field's tendency to overemphasize deficits—often at the expense of exploring more ecologically grounded hypotheses in developmental psychobiology (Carter et al. 2022; Nketia et al. 2021). Comparatively less academic attention has been paid to intersectionality among the research workforce, especially including neurodiversity and LGBTQ+ identities. Future considerations on how diversity impacts discovery will benefit from considering the contribution of intersectional identities.

2.2 | Diversity Among Participants and Partners

2.2.1 | Research Participant Diversity Is Necessary for Good Science

Ensuring participant diversity in our research is essential for the accuracy, generalizability, and ethical integrity of scientific findings (Nielsen et al., 2017). For example, the etiology and clinical presentation of heart attack symptoms in women have been updated numerous times over the last 20 years as research has included women and more accurately reflected women's symptoms (Jin et al. 2020). Unfortunately, a recent survey of the top five cognitive neuroscience journals revealed that the vast majority of manuscripts do not even report demographics of

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human participants (Goldfarb and Brown 2022). There is some good reason for this: chiefly, a concern for the long shadow of eugenics, which explicitly designated any racial or cultural differences as deficits relative to the dominant culture (Cardenas-Iniguez and Gonzalez 2024; Thomas et al. 2023). Both psychology and biology have a dark history and dominant role in the marginalization and abuse of people due to gender and sexual identity, ethnoracial identity, and disability status (Yakushko 2019). We should not be surprised that members of communities that have been experimented on, forcibly sterilized, and otherwise used for science as if they were not human beings but rather disposable subjects remain deeply skeptical of scientific institutions (Jaiswal and Halkitis 2019). And while increased diversity of the research workforce across these marginalized identities does contribute to increased participant representation, further work is needed to ensure culturally appropriate approaches (Roberts et al. 2020).

2.2.2 | Participatory Research Methods Aim to Address Opportunities for Harm

There do exist some models that attempt to address participant diversity and the ethical and human use of data from a justice framework. Situated neuroscience (Einstein 2012) and critical neuroscience (Slaby and Choudhury 2011) deploy techniques from ethnography and critical theory to contextualize the lived experiences of participants in studies, the positionality of the researcher relative to that population, and ethical issues of the work. While these techniques offer critical insights into the sociocultural context of data and researcher positionality, they do not, in themselves, transform structural or relational dynamics between researchers and communities. In contrast, participatory research methods try to materially change these dynamics and seek to diminish hierarchical relationships that can breed scientific negligence, mistreatment, and abuse (National Academy of Sciences 2009). In health research, participatory research methods were created to answer both practical and ethical concerns regarding the recruitment of minoritized or marginalized populations into research (Cornwall and Jewkes 1995). CBPR methods are particularly popular in health sciences. CBPR explicitly seeks to build equitable partnerships between scientists and community members surrounding a shared interest or concern. A meta-analysis on the impact of CBPR on enhancing ethnoracial diversity in clinical trials suggests that it is successful at both the recruitment and intervention levels (De Las Nueces et al. 2012). In the only published CBPR developmental psychobiological study we know of, the authors suggest that positionality maps and community advisory boards can help bridge the gap in understanding, make power dynamics more (though not completely) horizontal, and otherwise help in recruitment, implementation, quality, and impact of the work (La Scala et al. 2023). The twin goals of this approach are the betterment of the science *and* immediate benefit to the community.

3 | Democratizing Neuroscience for Rigor, Relevance, and Resilience

In a graduation speech for Spelman College, Princeton professor Ruha Benjamin said, "Black faces in high places will not save us" (Benjamin 2024). With this single quote, Dr. Benjamin captured a stance that is critical of representation-only movements, or movements that do not ask, "What are we being included into?" It emphasizes that diversity without systemic change merely changes the superficial, rather than the ultimate causes within the systems that make representation so difficult. This is highlighted in the current context by the reactions of universities to the anti-DEI opinion put forth by the current administration as if it were law. In a reversal of pro-DEI initiatives sparked around the racial justice movements in 2020, some universities are dismantling everything from scholarships aimed at increasing the participation of underrepresented groups to the more peripheral offices of belonging (e.g., Carter 2025). What does it mean to be included in a system that ultimately sees your participation as expendable? Beyond academia and in an unsettling echo of both Nazi Germany policies and the United States' own history of eugenics (Yakushko 2019), positions long discredited through scientific process and evidence-based consensus, such as mass institutionalization (Glenn 2025), anti-vaccine and anti-neurodiversity rhetoric (Neergaard and Stobbe 2025), ethnoracial genetic hierarchy (Tabery and Caplan 2024), and essentialist views on gender and sexuality (Johnson 2025) have resurfaced in political discourse. Developmental psychobiologists are uniquely positioned to dismantle these arguments, equipped with training and easy access to the research discrediting the foundations of these exclusionary positions. However, under the status quo of academia (e.g., lack of support from senior colleagues and admin, awareness, opportunity, or concern), researchers often struggle to communicate this knowledge outside the ivory tower—decreasing access for the public, who could use it to critically assess political rhetoric and make informed democratic decisions. And because it is the broader public who elect leadership and shape policy, expanding access to our tools and knowledge arguably strengthens democracy itself.

3.1 | What Is Science Democratization?

This current moment makes clear the inextricable link between science and democratic self-governance. Science democratization is a movement and philosophy built on this premise. Beyond science communication, science democratization prioritizes the sharing of power and responsibility between scientists and the lay public in the creation, dissemination, and application of science (Kurtulmuş 2022). Science democratization challenges the assumptions that science is a value-free enterprise—that is, a wholly impartial process—and that the public can reasonably judge the utility of science without understanding its process and products (Betz 2013). Instead, science democratization stands on considerable evidence suggesting the opposite: even using the scientific method, researchers' backgrounds influence the questions asked, the level at which evidence becomes acceptable, and the ways in which products make it to the public sphere (Frickel et al. 2010). Similarly, it is unreasonable to think that a public unschooled on the enterprise or process of science can assess its value independently, thus leaving them vulnerable to charlatans and despots (Sagan 1997), and to scientists themselves, who can wield unreasonable power with large blind spots. Therefore, because the scientist is only human and cannot reasonably keep track of all the ways in which their science could be expanded upon or (mis)applied, the scientist must rely on and engage with

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a diverse public with their own knowledge and expertise to guide their science towards meaningful science agendas and policy. Building trust between scientists and community members then becomes the linchpin of science democratization, and the cultivation of relationships and trust becomes the first goal of any scientist aiming to democratize science (Beran et al. 2021).

The process of democratization builds epistemic justice, or the right to access, understand, benefit from, and participate in the creation of knowledge and its benefits. Participatory methods like CBPR, but also citizen science (or science volunteerism), both attempt to build epistemic justice and highlight its utility for both science and the public. In contrast, current practices in academia—from education, training, research production, and dissemination—do not support epistemic justice or particularly distributive epistemic justice. Distributive justice is concerned with the equal distribution of epistemic justice regardless of social class and identities (Irzik and Kurtulmus 2024). For example, Black scientists' exclusion from NIH funding has largely been due to the epistemic injustice of devaluing knowledge on certain topics like health disparities (Hoppe et al. 2019). However, distributive epistemic justice does not necessarily track broader societal patterns of oppression. For example, the lack of research on osteoporosis in men results in underdiagnosis and is considered a form of epistemic injustice since men do not benefit as much from the science (Irzik and Kurtulmus 2024). Within the context of science democratization, who gets to do science, what kind of science gets done, and who gets to benefit from it all become inextricably linked with epistemic justice and ultimately the health of a democracy. We argue that approaching science democratization and epistemic justice as an ongoing practice strengthens democracy and betters science by improving rigor, relevance, and resilience.

3.2 | Science Democratization Improves the Rigor of Our Science

Rigor—defined as the reproducibility, validity, and accuracy of our findings—is an essential component of scientific inquiry (Prager et al. 2019). While developmental psychologists are trained to achieve rigor through experimental controls, adequate sample sizes, and matching methods to questions, science democratization highlights yet another source of rigor: the inclusion of diverse perspectives and relationships across the research process and the willingness to challenge or stress test the status quo. This includes expanding access for individuals from underrepresented groups to become developmental psychologists, diversifying study samples, and sharing decision-making and procedural power with nonacademic partners who hold their own unique expertise.

Democratizing science through the inclusion of underrepresented voices and dialogue expands a key part of rigor: thinking. There are numerous examples of Western cultural biases that were smuggled into the scientific literature without question—until the diversity of researchers was expanded to ask new questions. For example, for the majority of the past century, the orthodox view of birdsong was that males sang and, with a few rare exceptions, females did not. A growing number of women ecologists asked new questions, leading to the discovery that female birdsong is common and, in fact, the evolutionary ancestral trait (Odom et al. 2014; Haines et al. 2020). Similarly, while dominant social narratives have colored the interpretation of biological mechanisms, the inclusion of sociological and Black feminist perspectives draws out the biological realities of structural oppression and the often reasonable, if not consequence-free, adaptations to them (Dhamala et al. 2025; Nketia et al. 2021).

Similarly, science democratization via inclusive participatory research methods expands the ecological validity and the scope of our questions in developmental psychobiology. Psychology research broadly, and developmental psychology specifically, has a long history of challenges with limited study populations—with an overemphasis on Western, Educated, Industrialized, Rich, and Democratic (WEIRD) populations—that limits generalizability and applicability of findings to our broader diverse communities (Henrich et al. 2010; Majid and Levinson 2010; Oliveira and Baggs 2023; Clancy and Davis, 2019). The brain does not develop in a vacuum but in a specific context, where resources, identities, and challenges are intertwined. If we understand neural development as unfolding in an environment, then representation is a necessary aspect of understanding variations and the underlying brain–environment interactions that produce them (e.g., Garcini et al. 2022; Green et al. 2022; Nketia et al. 2021). In fact, a lack of representation can lead to vastly different conclusions about underlying mechanisms. For example, predictions on age-related changes to grey matter volume in children show a compelling pattern that could become a recommendation for pediatricians but fall apart once we account for the lack of representation in the data source (LeWinn et al. 2017). Practically speaking, engaging nonacademic partners and community members democratically does increase the likelihood of a more representative sample (Cornwall and Jewkes 1995). Less discussed is how active participants also expand the questions we ask and offer insights into data interpretation. For example, a community action board can reframe a theoretical problem in terms of its *meaning* and add context to findings (e.g., Brockman et al. 2021). Engaging in inclusion democratically, therefore, allows for increased rigor at multiple junctures of the research process.

3.3 | Science Democratization Improves the Relevance of Our Science

Science democratization goes beyond the walls of the academy—it improves relevance to the larger population. Relevance is enhanced because continued engagement with a community allows for the co-creation of research agendas that reflect real-world needs and lead to more refined questions with better translation practices. As we write this, scientists are calling for marches in defense of science. But why should a public who have been given little access to or understanding of science and scientists care to defend them? Research on trust in science suggests that people in the United States have had a generally favorable but vague understanding of science as a public good and have warned that vague speech from scientists about the public merit of science adds little value (Resnik 2011). Further, in an analysis of what prompted individuals to join the "March for Science," a demonstration against the first Trump administration's anti-science rhetoric in 2017, researchers found that both a sense

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of injustice and a sense of collective efficacy were important to increased intentions to advocate for science (Campbell et al. 2023). Along with many other scientific fields, developmental psychobiology and neuroscience face several hurdles to promoting public advocacy. First, since these disciplines are rarely encountered before a university curriculum and largely avoidable even then, epistemic justice is likely quite poor. Second, without general knowledge of what our science is and can do for the public, communities are at the mercy of both overly generous and overly negative interpretations of the science. Further, continued segregation from nonacademic communities can encourage poor communication practices, thus reinforcing a vicious cycle of isolation and perceived irrelevance. In contrast, inclusion and participation can create true relevance and meaning for those who participate and improve the resilience of our science and our citizens against the manipulation of information.

3.4 | Science Democratization Improves the Resilience of Science and Democracy

Finally, all of this is achieved while also creating greater resilience for science, public trust, and the public itself. For example, though reproducibility and open science practices are important to transparency, absent strong community trust, vehicles of transparency can facilitate harassment against scientists (Lewandowsky and Bishop 2016). Many of these harassment campaigns are politically motivated in the colloquial sense (e.g., motivated by party loyalty rather than an issue) and can be extreme (Nogrady 2024). Strong ties between communities and scientists open up room for nuance and resist sweeping generalizations about science and scientists from motivated parties. Further, access to scientists allows community members to enlist them in important work. For example, following the start of the COVID-19 pandemic, a longstanding research-community coalition was able to shift priorities to meet the needs of those they had been committed to serving (Nguyen et al. 2020). In this way, science democratization makes natural allies out of scientists and communities (see Figure 1).

3.5 | The Community Neuroscience Approach

The CNI has been developing a "community neuroscience" approach to science democratization. Our working definition of community neuroscience is that it is a practice of science democratization grounded in deep, sustained relationships between neuroscientists and community members. Rather than viewing community partnerships as a means to an end—such as participant recruitment or dissemination—community neuroscience treats being *in community* as an end in itself (Redman et al. 2021). We seek to make a joint sense of community first—to find a shared identity, mutual presence, and non-transactional connection—using the framework of learning communities (e.g., Stein and Imel 2002). Neuroscience becomes a shared activity and an important epistemic lens through which complex challenges are explored and collective goals are pursued. The CNI brings together researchers and local stakeholders in ways that blur the boundaries between science and civic life. Sometimes these gatherings involve structured research or educational programming. Other times, they simply involve showing up—at school reading days, community fairs, or advocacy events—out of a sense of care and belonging.

From this relational foundation, community neuroscience approaches democratization through three core principles: accessibility, translation, and inclusion. By accessibility, we mean that communities should have relatable access to the tools, knowledge, and people of neuroscience and psychobiology. This may mean the classic science dissemination for public consumption, but also more intimate gatherings like tours of fMRI facilities or creative events that encourage intertwining people's everyday interests with the lens of neuroscience. By translation, we mean that the community and the field of neuroscience should benefit from the co-creation of knowledge beyond academia, recognizing this as an epistemic and civic partnership across different forms of expertise. By inclusion, we mean that we affirm the necessity of diversity to epistemic distributive justice, to democracy, and to a vibrant, innovative, accurate, and ethical field (Swartz et al. 2019; Kurtulmuş 2022). It means rejecting the idea that inclusion requires community access and instead can look like training people from those communities and supporting them so *they* may expand our science and our public good.

This approach demands a mixture of grass–roots accountability and attention to structural barriers—cost, access, physical and cognitive accommodations—and seeks to build resilient networks of engagement. In a time of declining national support for inclusive research and science democratization, these local communities may offer the resilience of distributed systems for both researchers and communities. Below, we include more practical aspects of this approach using the CNI and a science democratization event as an illustrative example.

4 | Science Democratization in Practice: An Illustrative Event From Community Neuroscience

4.1 | CNI Governance and Priorities

The CNI is a science-democratizing entity that aims to build an ecosystem facilitating the democratization of neuroscience and psychobiology. Our network includes neuroscientists and psychobiologists working with humans and/or nonhuman animal models, researchers focused on neurodiversity and education, and community members who bridge neuroscience with real-world applications. While CNI shares many structural elements with an academic initiative—it has a faculty executive director, a graduate fellowship, faculty founders and advisors, and student and faculty volunteers—it also integrates nonacademic voices through an advisory board. This board comprises volunteers from industry and policy spaces who support community-engaged neuroscience and diversity initiatives. Their role includes facilitating network-building, offering organizational insights, and participating in events. For example, at the event described below, Dr. Maureen Dunne—a policy leader in neurodiversity—helped ground our discussions in real-world issues such as how companies decide policies on whether and how they will support neurodivergent workers.

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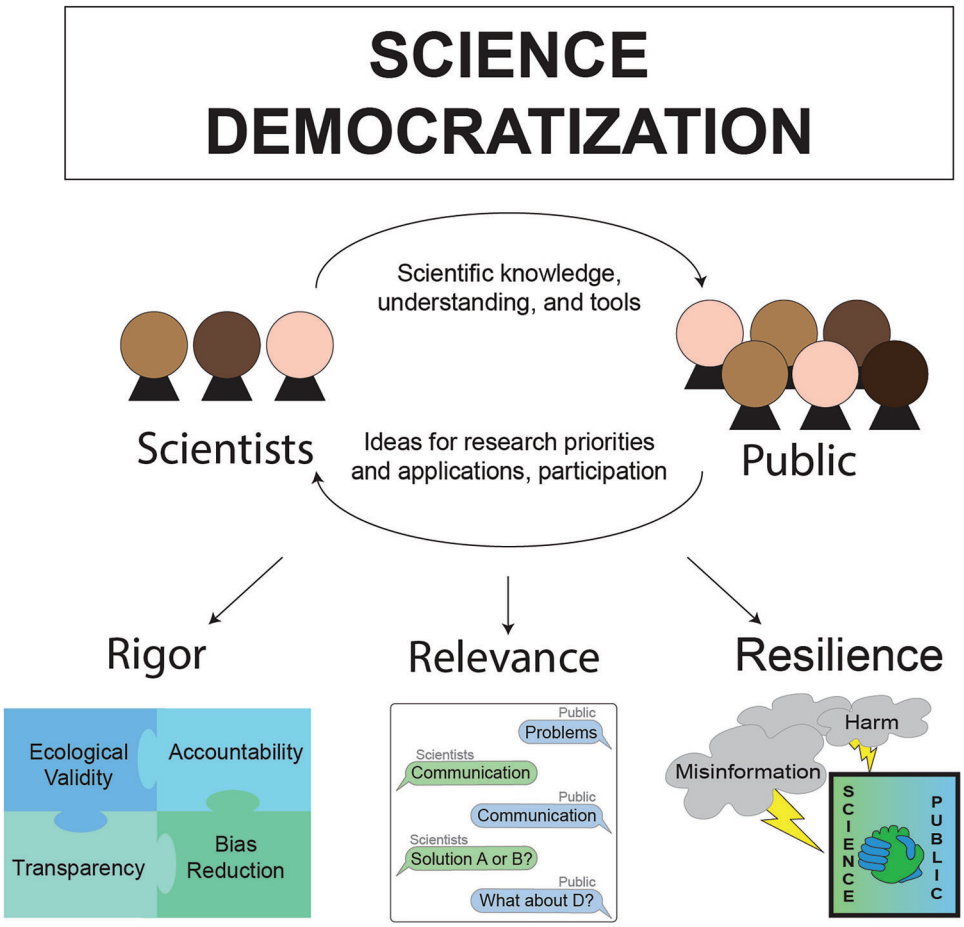


FIGURE 1 | Impact of science democracy on science-community partnership, scientific rigor, relevance, and resilience against anti-democratic attacks.

Though CNI partners with university-embedded programs like Upward Bound, much of its work takes place outside university walls. Our primary focus is Syracuse, NY, located about an hour north of Cornell University. Syracuse has one of the highest childhood poverty rates for a city of its size, a diverse ethno-racial composition (US Census Bureau 2020), and a strong network of educators, healthcare professionals, and community organizers committed to local well-being. This combination of need and strength makes it an ideal site for community neuroscience. CNI predominantly creates events that foster diverse stakeholder engagement in a neuroscience topic relevant to our community audience. For example, in our Accessible Neuroscience Speaker Series, neuroscientists prepare lectures on sleep or healthy aging for an audience of teachers and caregivers. Fundamentally, CNI does not exist to simply disseminate knowledge from experts to the public. Instead, we build knowledge collectively, ensuring that both neuroscientists and nonacademic community members shape the direction of research, science communication, and practical application. For example, we ask community members what they would like emphasized in our speaker series, both in terms of topics (e.g., learning, sleep) and discussion questions (e.g., relationship to socio-emotional learning goals). In this way, the initiative is not only about neuroscience—it is about rethinking the very structures that govern how knowledge is produced and who has the authority to produce it.

4.2 | Community Neuroscience Celebration

A key objective of CNI's democratizing events is to increase epistemic justice by expanding who gets to participate in neuroscience and how knowledge is shared. One such open event, the Community Neuroscience Celebration: Honoring the Impact of Women and Nonbinary Leaders event, took place in May 2024 with support from the President's Council for Cornell Women (PCCW). This event was not just a symposium or a networking opportunity: it was an intentional effort to democratize neuroscience by celebrating the contributions of women and nonbinary scientists while also ensuring that those outside of academia could meaningfully participate in the conversation. The intended audience included national policy figures using neuroscience in advocacy, local upstate New York researchers, neuroscience trainees, administrators, and community partners, particularly educators from My Brother's Keeper Alliance Teacher Opportunity Corp II, an initiative promoting underrepresented identities in teaching (see Table 1). Rather than hosting the event at the university, we chose a third space in Syracuse, NY—a neutral, community-centered location—ensuring accessibility for both traveling scholars and local community members. This also addressed power dynamics that often arise in university-centered engagement. Travel expenses, childcare accommodations, and food were covered to remove participation barriers.

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TABLE 1 | Community Neuroscience Initiative Celebration Event Participation by Sector.

Sector	Number of participants
K-12 Educator and administration	2
Science consulting and policy	4
Higher education and administration	4
Research and/or teaching trainees	15
Research staff	5

This section describes a specific event, participants' experiences, and the authors' personal reflections framed alongside literature on learning communities and barriers to science participation. While not presented as formal qualitative research, it offers an illustrative account of science democratization in practice—insightful, grounded in experience, and rarely documented in our fields.

4.2.1 | Gathering Our People

Before we invited graduate trainees to participate in the event, our priority was identifying nonacademics to lead the professional development process. We began looking inward at our established collaborations with educators, nonprofit leaders, and philanthropic advisors who were stakeholders in enhancing the representation and influence of women and nonbinary leaders in science. These informal networks were sustained through ongoing partnerships—whether it was showing up to read to young children, supporting initiatives to make the teaching profession more accessible, engaging in neurodiversity advocacy, or simply through continuing conversations about an issue or shared goal. We reached out with a personal invitation to help us train graduate students to listen to community members. Though the "official" invitation was through emails and we had a registration portal, the real invitations were often in person, through text, or in a one-on-one meeting, which allowed for clarification and collaboration. We made it clear that our goal was for scientists to learn from the community—and that trainees would receive their professional development directly from community members, not just about them. Notably, while most trainees were fairly new to CNI, the majority of nonacademic attendees were long-term community partners who participated in previous CNI programming, underscoring the importance of sustained partnership. In a very real sense, they showed up to help us train the trainees.

With this conceptualization in mind, we also sought to invite professionals in and outside of academia who work at the intersection of public policy, social practice, and neuroscience. Two individuals were already CNI partners—one a board member (Dr. Maureen Dunne) and the other a previous presenter at our accessible neuroscience lectures (Dr. Sarà King) with a background in leading mindfulness practices. However, we also wanted to use this opportunity to attract someone new who was more closely connected to public policy and government. We therefore invited Dr. Frances Colon, someone with a PhD in neuroscience who chose to use this background to become first a government science advisor and then an international public policy expert on climate change. She also shared her work in helping individuals from diverse backgrounds enter government service.

Finally, we invited graduate and postgraduate students from across the region. Since the event would take place in the central lakes area of New York, we looked to the surrounding schools and departments of psychology, neuroscience, cognitive science, and biology. We shared the event information and registration portal with departments, who would email it out to others. In the end, the venue was attended by trainees and faculty from Cornell University, Syracuse University, the University of Rochester, and SUNY Upstate Medical, representing the regional diversity and their different experiences in collaborating beyond academia.

4.2.2 | Program Activities: Breaking Down Traditional Hierarchy

Tables 2 and 3 show the daily programs at a glance. From this view, the event resembled a small symposium with keynote talks, student flash talks, and awards. However, two features set it apart: (1) the use of interactive learning community methods in keynotes and (2) setting nonacademic participants as experts in their own right.

Keynote presentations did not function as top–down lectures but instead incorporated mindfulness exercises and Think-Pair-Share activities aimed at reinforcing psychological safety and broadening participation (Mundelsee and Jurkowski 2021). The event began with attendees co-creating definitions of "democratizing neuroscience" and "community neuroscience" as well as establishing collective aspirations for the event. The facilitator (the CNI executive director) instructed participants to do a Think-Pair-Share. Participants first focused on their own thoughts without referencing some perceived "right answer." Then, they shared and co-constructed the definitions again with a partner before finally sharing with the broader team. Using reflective listening and summarizing techniques, the facilitator's goal was to start the program on a foundation of mutual understanding, shared ownership, and equity. By engaging all attendees in the process, the goal was to ensure that these definitions did not privilege one perspective over another.

To underscore the community as diverse experts across ages and sectors, the community panel featured an elementary school principal, a teacher-in-training, and a foundation leader (the

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TABLE 2 | CNI Celebration Program Agenda Day 1—Sunday, May 19, 2024.

Time	Session	Speaker
8:30 a.m.	Welcome and Introduction: Attendees co-created definitions of "democratizing neuroscience" and "community neuroscience." This practice was introduced to build the program on a foundation of mutual understanding, shared ownership, and equity. By engaging all attendees in the process, the goal was to ensure that these definitions did not privilege one perspective over another. Community aspirations for the retreat were set.	Dr. Marlen Gonzalez
9:45 a.m.	The Science of Social Justice Framework: A Cultured Embodied Movement: Attendees learned about exploring community well-being and social justice through the intersection of art, identity, and technology.	Dr. Sarà King
11:00 a.m.	Lunch and Networking Break	
12:15 p.m.	Community Award, Neuroscience Luminary Award Lecture: Dr. Colón's journey of becoming deputy science and technology adviser to the Secretary of State during the Obama administration, focusing on the global advancement of women in science and climate policy.	Dr. Frances Colón
1:30 p.m.	Community Panel Discussion: A panel of community leaders and trainees discussed what neuroscientists need from communities and what communities need from neuroscientists. Moderator: Dr. Marissa Rice	Principal Kuricheses Alexander, Crystal Nunez
2:30 p.m.	Networking Break	
3:30 p.m.	The Brain Skin Connection: Research on Your Largest Endocrine Organ: A CNI-accessible neuroscience lecture on the biology and ecological influences on skin. Attendees learned about Dr. Aguirre's journey as a science and business leader.	Dr. Claudia Aguirre
4:30 p.m.	Dinner and Reflection	

Note: This table provides an overview of the program activities from the first day of the retreat. The schedule can be viewed at https://sites.google.com/cornell.edu/cni-leadership-celebration/schedule.

TABLE 3 | CNI Celebration Program Agenda Day 2—Monday, May 20, 2024.

Time	Session	Speaker
8:30 a.m.	The Diverse Brain: Championing Inclusivity in Higher Education: Attendees discussed what neurodiversity and inclusion really mean.	Dr. Eve DeRosa
9:45 a.m.	Embracing Neurodiversity in Career Development and Research Thinking: Attendees learned about the writing process and translational impact of *The Neurodiversity Edge*, in addition to the importance of community neuroscience.	Dr. Maureen Dunne
11:00 a.m.	Lunch and Networking Break	
12:15 p.m.	Student and Trainee Flash Talks	Grad Students and Trainees
1:30 p.m.	Community Neuroscience Luminary Emerging Scholar Award Lecture: Dr. Riley shared her experiences working with communities at the VA Hospital in Boston and supporting diverse community recruitment at Cornell. The talk concluded with a Q&A focused on women's leadership topics.	Dr. Elizabeth Riley
2:30 p.m.	Networking break	
3:30 p.m.	Consultancy: Trainees and community members received a description of a proposed study exploring how single and partnered mothers use their social support networks to manage their stress. They were also informed of an opportunity to work with a community partner serving low-income families. A problem and question were then presented for the consultancy: Given that mothers have limited time and money, what gifts and experiences might be valuable to first-time mothers that can be provided now for their participation in the study? The group brainstormed ideas like childcare, providing education outside of the study, ensuring that the mothers and infants feel safe, and listening to their concerns and needs.	Everyone
4:30 p.m.	Dinner and Reflection	

Note: This table provides an overview of the program activities from the second day of the retreat. The schedule can be viewed at https://sites.google.com/cornell.edu/cni-leadership-celebration/schedule.

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TABLE 4 | Community Panel Discussion Questions.

Question
1. In your work, how do you see yourself involved in neuroscience?
2. How do you interact, or how do you wish you interacted, with neuroscientists?
3. Why did you decide to get involved with the neuroscientists at the CNI?
4. What should neuroscientists interested in health, education, or translation know about your work with communities?
5. What do you think is an ambitious goal neuroscientists should set in building relationships with communities, and how should we achieve it?

Note: This table lists the questions that were posed to a panel of community educators and trainees during the first day of the retreat.

latter unable to attend due to a family emergency). The panel was moderated by a graduate of Cornell's psychology department in the neuroscience area. The panel members discussed their interactions with neuroscience, what they needed from neuroscientists, and how scientists should approach relationship-building with communities (see overarching questions in Table 4). Next, trainees gave 10-min-long flash talks on their work. This allowed them to practice communicating their work simultaneously to researchers, stakeholders, and community members with the intent of strengthening their capacity to engage diverse audiences. Then, trainees fielded questions from this audience. Of note, perhaps because this was done on the second day, there was an ease in this process and open dialogue that ranged from community members giving alternative hypotheses to simple communication of excitement for the person's research.

Perhaps the most explicit act of epistemic justice occurred during the *consultancy session*, which paired neuroscientists with community members to engage in structured problem-solving. Rather than positioning neuroscientists as the default experts, the consultancy validated nonacademic expertise by giving equal weight to both perspectives. The consultancy session used a structured collaborative problem-solving framework commonly found in learning communities and community-engaged organizations (e.g., Civic Canopy Consultancy Protocol). Short descriptions of two real challenges needing feedback were prepared before the event in conjunction with the CNI team. One, from a neuroscientist, involved increasing participatory elements in a study with new mothers and infants; the other, from an educator, involved communicating the importance of sleep to elementary school parents.

Consultancies typically involve a host (who presents the dilemma), a facilitator (who guides discussion and enforces participation rules), a timekeeper, and a group of consultants. Originally, the researcher and educator were to alternate as facilitator and host. However, due to time constraints, we invited the educator to choose her role and the challenge to address. She opted to facilitate the research-focused consultancy. As facilitator, she led the session, enforced protocol (e.g., limiting the host from speaking during feedback), and helped center community concerns and desires. Together, these activities reinforced shared purpose and reciprocal expertise, ensuring that both neuroscientists and educators were valued knowledge contributors.

4.2.3 | Impacts: Epistemic Justice in Action

In preparation for reporting on this event, we asked attendees to reflect on their experience by going through their notes from the events, engaging in group communication, and providing updates on post-event activities and impacts. Several key themes and insights emerged from individual meetings, group debriefs, and a short-answer questionnaire. Though this is not a qualitative research study, the community between psychologists and nonacademics remains rare, making it important to document and share lessons from this event.

4.2.4 | What Drew People to the Event?

Attendees cited shared interests in science communication, community engagement, and the event's emphasis on celebration as major motivations for attending. Researchers across career stages saw this as an opportunity to affirm the importance of outreach and engagement; early-career researchers in particular sought a sense of belonging and validation for their interests in community-based work. Educators and nonacademic attendees were drawn to the event by both their preexisting relationships with CNI and the chance to interact with women and nonbinary neuroscientists. Keynote speakers were drawn in by the connection to policy and interest in dismantling barriers between neuroscience and the community.

However, several barriers to participation emerged. Participants cited transportation as a potential deterrent, which was largely resolved through carpooling. Childcare responsibilities were a significant hurdle, reflecting systemic barriers that disproportionately impact women in professional spaces (Calisi and a Working Group of Mothers in Science 2018; Machlovi et al. 2021). Notably, securing funding for childcare was more challenging than for travel, highlighting how structural definitions of "reasonable costs" exclude caregivers from professional development. Work commitments also shaped attendance. While some teachers could only participate in the Sunday session, one graduate trainee hesitated to attend due to concerns that their Principal Investigator (PI) might not approve of them "taking time off" for a community-oriented event. In contrast, every other trainee attendee shared that their PI encouraged their attendance. This contrast underscores how academic culture can either facilitate or hinder participation in democratized science, depending on whether mentors actively support these efforts.

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4.2.5 | Immediate Impacts: Expanding Knowledge and Fostering Belonging

Nearly all attendees initially expressed some level of uncertainty or anxiety—many were concerned about whether they would have meaningful contributions in such an interdisciplinary and interprofessional space. However, participatory activities, such as Think-Pair-Share discussions and mindfulness exercises, helped facilitate dialogue and ease discomfort. Attendees reported that these exercises, along with shared meals and informal discussions, strengthened a sense of belonging and confidence in their expertise.

Research trainees and educators especially described interactions with each other as profound. Trainees described educators as role models, particularly in their ability to apply neuroscience concepts to real-world challenges. Educators provided critical feedback on how neuroscience could better serve their communities. One teacher, for example, discussed how a previous neuroscience talk reshaped her approach to student misbehavior, encouraging and practicing deep breathing exercises to promote self-regulation. Another shared their ongoing efforts to secure wobble chairs, fidget toys, and study supply packs to support both neurodiverse and economically disadvantaged students, demonstrating how neuroscience-informed approaches could enhance inclusivity in classrooms.

Keynote talks on neurodiversity, community engagement, and science policy opened space for conversations that many attendees had not encountered in traditional academic or professional settings. For some, this was the first time they had reflected on how academia affects their mental health, how gifted/remedial tracking shaped their educational experiences, or how systemic barriers affected them as mothers working in neuroscience research. Many attendees felt that these discussions helped them contextualize their own struggles, reinforcing the need for structural changes in academic and scientific spaces (Hoppe et al. 2019; Matias et al. 2022).

4.2.6 | Post-Event Impacts: Changes in Perceptions and New Collaborations

In follow-ups, participants described lasting changes in self-perception, professional outlook, and academic cultural awareness. Many reported an increased sense of purpose, motivation, and responsibility to apply what they learned. Research trainees, in particular, described greater awareness of how scientific communication can be unintentionally exclusionary and a commitment to making their work more accessible.

Some attendees found that the event shaped how they evaluated other academic meetings and values. One trainee, for instance, described feeling a sense of disconnection at a traditional conference after the Celebration, realizing how absent broader societal engagement and collaborative spirit were by comparison. Another human-research trainee reported that the event gave them language to articulate their dissatisfaction with academic spaces that fail to engage with public needs or concerns. A student in basic nonhuman animal research reflected on how discussing results with the community might lead to setting better priorities for their field. Conversations sparked by the event led attendees to ask the fundamental community neuroscience question: What can neuroscience do for the community, anyway? Even those without immediate plans to act on this question viewed it as a guiding principle for their future training and career decisions. One trainee stated, "I started thinking about who I want to be in my community and how I want to give back, just as mentors in CNI gave to me." Further quotes can be found in Figure 2.

Beyond personal transformations, the event catalyzed several unexpected collaborations:

- One educator recruited colleagues to attend future neuroscience outreach lectures and explore a research project related to education.
- Another partnered with a researcher to apply for funding for a community-centered neuroscience project.
- CNI was invited to a neurodiversity fair hosted by a Syracuse public school coalition.
- One researcher refined her dissertation plans based on insights gained from the consultancy.
- A keynote speaker joined the CNI advisory board, reinforcing long-term engagement.
- Several trainees made efforts to bring neuroscience knowledge to their home communities—one working on foreign language skills to communicate neuroscience to non-English-speaking family members, and another attempting (but ultimately being blocked by national policies) to speak at her former high school outside of the US about neuroscience careers. The former high school only allowed officially approved organizations to conduct events, preventing individual-led workshops despite some excitement for the work.

The continuing relationships among participants illustrate how democratized science creates lasting, organic collaborations rather than transactional interactions. Many remain in contact, supporting each other's work, and this very manuscript is a product of those sustained connections.

4.2.7 | Organizers' Reflections and Lessons Learned

Post-event reflection is a crucial component of democratizing and participatory science approaches. As organizers, we created this event with the intention of providing others with the kind of community we wished we had earlier in our academic journeys. While we anticipated meaningful exchanges, we were moved by the depth of engagement and the enthusiasm of participants. The event affirmed many of our beliefs about the power of shared spaces for community neuroscience while also revealing areas where we can improve in future iterations.

One of the clearest successes of this event was how it fostered relationships that extended beyond the formal programming, reinforcing that neuroscience is most impactful when embedded in reciprocal, community-centered dialogue. Attendees found

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FIGURE 2 | Quotes from undergraduates, graduate students, and postdoctoral scholars about the barriers they experienced trying to attend the event, the feeling of celebration that brought them together, and their reflections on how community neuroscience has spurred them to democratize science.

new collaborators, strengthened their sense of purpose, and engaged in knowledge-sharing that was both rigorous and accessible. However, while this event successfully redistributed knowledge authority, some traditional hierarchies still emerged—particularly in the relationship between organizers and trainees. While trainees were deeply engaged participants, their role in shaping the event could have been expanded. Greater involvement from trainees and community members in the event design process—from selecting themes to structuring discussions—would have strengthened our commitment to shared power and responsibility. Further, though we concentrated on inviting individuals with whom we had a strong *working* relationship, future events could further include other individuals who attend our dissemination events, such as parents and patients. Beyond program structure, we were reminded that seemingly small logistical decisions—such as food—play a crucial role in fostering community. We intentionally selected local, well-loved food venues, turning mealtimes into opportunities for deeper conversation. Providing food containers for attendees to take meals home was a small but meaningful way to prioritize accessibility, care, and hospitality. While often overlooked in academic spaces, these decisions contribute to a more inclusive and welcoming environment.

We also reaffirmed that events like these require significant social capital and institutional support. The success of this event was due in large part to the concentrated efforts of a small organizing team, including one professor, one graduate student, a team of undergraduates, and faculty and professional volunteers. Logistical coordination—including securing funding, coordinating reimbursements, and managing communications—was made possible through the support of the BCTR. Similarly, participation from educators was strengthened by the encouragement and support of their institutional leaders. These elements were crucial in ensuring broad participation, but they also highlight a challenge: democratized science should not rely solely on informal networks and volunteer labor. Future efforts should explore sustainable funding models and institutional commitments that can better support community-based neuroscience engagement. Moreover, we acknowledge that there are multiple pathways to science democratization, including through "third sectors" (Domènech 2017; see Hardwick et al. 2015), which may be natural allies for our efforts.

As we look ahead, we recognize that moving from diversity to democratization requires continuous reflection and adjustment. Based on our reflections, the next step is to deepen community participation in all stages of event planning, research design, and evaluation. By doing so, we move beyond simply including more voices and instead work toward creating scientific spaces that are truly co-owned by all participants. We wish these next steps in community neuroscience had been captured by rigorous qualitative research. While informal feedback and post-event reflections provided invaluable insights, our data collection could have been more systematic. Inviting a qualitative researcher to participate from the outset—both as an observer and as a collaborator—would have ensured a richer, more structured evaluation of our impact. We applied this lesson to other CNI activities and hope to provide a more systematic analysis in the future.

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TABLE 5 | Reading Recommendations.

Focus area	Recommended readings
Ethical and philosophical frameworks	Escott, S., R. Cullen, and Q. Abraham. 2024. "Global Psychology, Solidarity and Social Justice." *Educational Psychology Research and Practice* 10, no. 1, 1–13. https://doi.org/10.15123/uel.8yv8z. Comas-Díaz, L., and E. Torres Rivera, eds. 2020. *Liberation psychology: Theory, method, practice, and social justice*. 17–40. American Psychological Association. https://doi.org/10.1037/0000198-001 Malherbe, N., and H. Canham. 2024. "Reading a liberation psychology archive in South Africa." *South African Journal of Psychology*, 54, no. 4: 475–487. https://doi.org/10.1177/00812463241267295.
Examples of challenging science epistemology	Duarte, N., L. Massi, and L. A. Teixeira. 2022. "The Committed Objectivity of Science and the Importance of Scientific Knowledge in Ethical and Political Education." *Science & Education* 31, no. 6: 1629–1649. https://doi.org/10.1007/s11191-021-00302-2. Diab, M., Y. A. Jamei, A. Kagee, and G. Veronese. 2020. "The interplay of paradigms: Decolonizing a psychology curriculum in the context of the siege of Gaza." *Nordic Psychology* 72, no. 3: 183–198. https://doi.org/10.1080/19012276.2019.1675087. Nxumalo, F., and S. Cedillo. 2017. "Decolonizing place in early childhood studies: Thinking with Indigenous onto-epistemologies and Black feminist geographies." *Global Studies of Childhood* 7, no. 2, 99–112. https://doi.org/10.1177/2043610617703831. Wijngaarden, V., and P. N. Ole Murero. 2023. "*Osotua* and decolonizing the academe: implications of a Maasai concept." Supplement, *Curriculum Perspectives* 43, no. S1: 33-46. https://doi.org/10.1007/s41297-023-00190-2.
Experience and applicable lessons from grassroots and action research movements	Liboiron, M. 2016. "Care and Solidarity are Conditions for Interventionist Research." *Engaging Science, Technology, and Society* 2: 67–72. https://doi.org/10.17351/ests2016.85. Lykes, M. B., and E. Sibley. 2014. "Liberation Psychology and Pragmatic solidarity: North–South Collaborations Through the Ignacio Martín-Baró fund." *Peace and Conflict: Journal of Peace Psychology* 20, no. 3: 209-226. https://doi.org/10.1037/pac0000045. Toenders, Y. J., K. H. Green, L. W. te Brinke, R. van der Cruijsen, S. van de Groep, and E. A. Crone. 2024. "From Developmental Neuroscience to Policy: A Novel Framework Based on Participatory Research." *Developmental Cognitive Neuroscience*, 67: 101398. https://doi.org/10.1016/j.dcn.2024.101398.

Note: This table shows a list of recommended readings to help deepen the understanding and practice of creating a democratic structure for researcher-community member relationships.

4.3 | An Invitation to Practice

Thus far, this paper has been motivational and demonstrative, but not instructional. This is on purpose, as we would like folks to engage in practice and questioning rather than in making a recipe for which they may not have the same ingredients. A democratic approach is inherently relational, and thus the specific practices that lead to participatory engagement will also be as heterogeneous as the types of relationships we can engage in. For example, common CBPR practices, like a community action board made of stakeholders (La Scala et al. 2023), could be essential in some contexts and projects and work to create relationships. Yet, it may feel performative or distancing in groups that already share regular, informal contact. Many of the skills needed are better learned by helping set up a Saturday pancake breakfast than in any graduate program. For example, the logistics of getting space or coordinating schedules, or the reflective listening skills to hear and understand concerns without being defensive. Further, science democratization is a value that acts more like an adverb. It describes how we do things rather than *exactly* what we do. We seek to do our science more democratically in the same way that some may do science more critically. This fundamentally changes decision-making and goal-setting but is a road without an end.

Nevertheless, there are next steps on any road. We offer two resources: a reading list (Table 5) and suggestions for praxis/practice (Table 6). In terms of readings, we want to acknowledge that there is quite a lot of literature and books on participatory and action-research methods, which do provide methods for conducting science more democratically. However, we think that some of the most necessary readings are not intentionally instructive but describe or reflect the status quo of the relationship between science and society or between scientists and the community. We believe that a truly democratic mindset has the following essential quality: before you consider community members to be participants, informants, experts, or co-creators, you simply choose to care about them, and the degree to which they advance your science is inconsequential. The notion of researchers as community members—or, at the very least, partners—has been written about academically as "solidarity" across a number of fields (e.g., Dawson and Jennings 2012; Sareen et al. 2023). This concept has no single operationalization but ultimately asks us to, colloquially, show up for people (Ross et al. 2023). These readings will lead to questions about ethics, about the actual business of science and its contradictions, and about the real harm that believing our epistemology needs no others has done (Hajir and Qato 2025; Veronese and Kagee 2024).

We recognize that this stance—choosing to care first—may seem unprofessional within psychobiology and neuroscience, where objectivity is often equated with detachment. But as Nejadmehr (2017) shows, this view is rooted in a scientific tradition shaped by

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TABLE 6 | Pitfalls, questions, and actions towards science democratization.

Pitfalls to avoid	Questions to guide self-reflection[a]	Examples of democratizing actions
Avoid equating knowledge of liberatory approaches with actual situated self-reflection	How do the intersectional identities[b] and institutions I represent—and the history of this community with those identities—influence our relationships and power dynamics?	Participate in valued community activities in ways that enhance liberatory practices, for example, challenging historical and current oppression.
	Has my field or institution inflicted harm on this community?	Acknowledge past harms and collaborate on proactively avoiding new harms.
	Because of my identities, what power and knowledge am I perceived to have, and how might this shape my interactions with this community?	Regularly examine your actions and relationships to ensure that your well-intended actions do not inadvertently sustain power relations.
Avoid assuming the superiority of your knowledge and your capacities	Are my skills applicable to the issue at hand, and are they the method the community wants to work with?	Prioritize the democratic process beyond the outcome or goal. *How* is just as important as the *what*.
	Does my training exclude, ignore, or deny local methods to deal with the issue at hand?	Resist unilaterally offering solutions and focus on co-creating solutions with partners.
	What technical skills do I have to offer, and how do they relate to the interests of the community?	Remember, you are just one expert among a community with diverse expertise, and your knowledge is not inherently more important.
Avoid valuing products over relationships	Am I aiming for epistemic justice by assigning equal value to all perspectives involved? What knowledge do I lack?	Build solidarity by cultivating non-transactional frameworks, prioritizing mutual care before even coproduction.
	Do I show up with curiosity and respect for the contextual knowledge the community has about the issue?	Value community priorities and move beyond "evidence-based" by contextualizing what matters and to whom.
	What knowledge about the issue derives from lived experiences that I have not had?	Build trust and rapport for the sake of trust and mutual understanding, not as a means to an end.
Avoid objectification, even ones that seem benign	How do I and my team interact with community members? How often are we leading versus following conversations? Where do we physically situate ourselves?	Engage in activities that build horizontal teamwork—learning *with* partners, not just *about* them.
	How do the conversations we have as a research team contribute to or push against objectification?	Prioritize community-generated questions and especially informal, spontaneous conversations.
	Do I avoid voicing disagreements with partners out of genuine respect—or am I being patronizing, assuming they can't engage in dialogue and resolution as equals?	Find what environments facilitate open, safe exchanges and create more of those.

[a]These are written to reflect on a specific situated community, not just a broad social category (e.g., the LGBTQ+ community in a specific community center versus the LGBTQ+ community generally).

[b]Consider a range of identities (e.g., race/cultural/age/gender/economic background/ability status/citizenship status).

colonial and racial hierarchies, where care was seen as a threat to authority. Neuroscience inherits this legacy, and as discussed earlier, overreliance on WEIRD samples and decontextualized methods has compromised both precision and inclusion. If care is seen as antithetical to our science, we must ask: can our science still be ethical? Duarte et al. (2022) offer a path forward through "committed objectivity"—a model that embraces science's fallibility and social embeddedness as a condition of *rigor*. Democratizing neuroscience begins by rejecting the false choice between objectivity and care and recognizing that care is not a weakness but a foundation for ethical inquiry. Indeed, we hope that the readers find the idea of relinquishing care not only misguided but also intolerable.

While the readings help illuminate the values and structural critiques behind science democratization, the practices in Table 6 offer ways of inhabiting those values—however imperfectly—in real contexts. In other words, reading invites reflection; praxis invites relation. Beyond reading, taking part in community events organized by community members is a key way of understanding your place in this shared space—or realizing when that space is not yours to enter but to support from the margins. If care is the center of a democratizing science wheel, then genuine curiosity and an almost hyperlocal attentiveness are its spokes. Working democratically requires ongoing, situated inquiry (Tomkins and Bristow 2021). You may understand how anti-trans legislation is playing out nationally, but do you understand how it has shaped

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local experiences and relationships? You may feel a sense of belonging with the people you hope to work alongside, but do you understand how the *institution* you represent might complicate that perception? Working democratically asks us for a level of precision and reflection that is, in a word, rigorous.

The resources we offer, both readings and praxis, are not gospel but guideposts—shared not as doctrine but as an offering from one group of travelers to another. Readers may find themselves in disagreement with some of the materials. That is part of the point. The goal is not consensus or purity, but movement: toward plurality, toward deeper thinking, and ultimately, toward practices that make democratization a lived—not just conceptual—part of developmental psychobiology.

4.3.1 | The Elephant in the Room

We offer these reflections in the hope of progress toward science democratization within academia. But it must be said: the academic institutions in which we do this work are not neutral bases of operations. While we strive for science democratization, many universities remain financially and academically entangled in global harms—including what has been described as scholasticide in Gaza and the suppression of dissenting academic and student voices (Hajir and Qato 2025; Veronese and Kagee 2024). These entanglements expose a deeper contradiction: institutions that publicly champion equity and democratization often fail to align those values with their economic and geopolitical practices. That misalignment places limits on our ethics and what science democratization can achieve within its walls. Even our efforts to "democratize" science risk reinforcing extractive dynamics when housed within systems that commercialize knowledge and concentrate power (Small and Mallon 2007; Tuck and Yang 2012). This contradiction cannot be solved within the bounds of the institution alone. We must ask not only what practices make science more democratic, but also whether our institutions are structured to make that possible—and at what cost elsewhere. In some cases, this may mean expanding beyond institutional categories altogether into third spaces or alternative infrastructures (Domènech 2017). In all cases, it means beginning not with outputs or deliverables, but with care (Liboiron 2016). What we offer is not more important than the conditions under which it is offered—or the people to whom it is offered.

5 | Conclusion

The role of developmental psychobiology (along with neuroscience) in shaping public policy carries both promise and peril. The potential to contribute to social good is significant, yet historical and contemporary examples demonstrate that without intentional safeguards, diverse participation, and community engagement, we can perpetuate harm. Further, despite the positive impact of diversity on our science, institutional commitments to DEI have proven fragile in the face of political pressures, and forced conditions of scarcity can make us all afraid. This paper argued that science democratization provides a philosophical stance that solidifies our commitments to DEI across the workforce and participants, as well as helps us improve public trust in science (Kurtulmuş 2022). Further, we have outlined how democratization enhances the rigor, relevance, and resilience of our fields by providing an avenue for community engagement in research questions, methods, and applications.

The CNI and community neuroscience provide one model for practicing science democratization. Through events like the Community Neuroscience Celebration, we have seen how structured spaces for reciprocal engagement can build trust, foster collaborations, and create conditions for knowledge-sharing that extend beyond the traditional boundaries of academia. To be clear, community neuroscience is not just about hosting events; it's about creating an ecosystem that can nurture new visions, ideas, and partnerships to reshape the business of science. It is a work in progress. As the political landscape shifts, scientists and communities must stand together to defend each other. Epistemic justice is both a goal and a tool to help us persevere and even transform (Irzik and Kurtulmus 2024).

We recognize that the democratization of psychobiology puts researchers in a position most of us were not trained to be in: that of a democratic advocate. Institutions may not support what happens beyond science communication and perhaps some degree of participatory design. This can impact promotion, especially that of faculty from underrepresented groups (Hoppe et al. 2019; Matias et al. 2022). Undeniably, those who are most vulnerable do need to find ways to garner the formal recognition, funding, and career incentives we need to continue our work. Yet there is value in engaging with communities in ways that may never translate into citations, grants, or promotions but that will strengthen the social fabric necessary for science to thrive in uncertain times. And if our anecdotes from CNI are generalizable, the engagements create a sense of purpose and belonging—intangibles that contribute to our well-being and perseverance. Even less formal forms of engagement, by providing consults for advocacy groups or engaging in local youth-serving organizations, can contribute to scientific trust and well-being. Less vulnerable individuals in leadership positions will need to do more than this and provide examples of commitment to epistemic justice. At the risk of engaging in stolen valor, we echo Toni Morrison's call to action in her address to members of the press (Morrison 2019, 40) and implore our research community:

"Move us closer to participatory democracy, help us distinguish between pseudo experience and a living one, between an encounter and an engagement, between theme and life. Help us try to figure out what it means to be human in the twenty-first century."

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Conflicts of Interest

The authors declare no conflict of interest.

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Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

Endnotes

[1]While we note that neuroscience and psychobiology have different histories and traditions, for the purposes of this article, we will use them somewhat interchangeably for ease of communication.

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